Evaluating NLI Models under Disagreement: DeBERTa-v3 and ModernBERT on ChaosNLI

Dujana Abrar Jiayin Feng

Abstract

Natural language inference (NLI) benchmarks usually keep one gold label per example, even though people often disagree. ChaosNLI (Collective Human Opinions in oft-used NLI evaluation sets) collected 100 labels for 1,514 SNLI and 1,599 MNLI-matched items and scores models against that human distribution as well as against the majority vote. Prior work showed that BERT, RoBERTa, XLNet, BART, and ALBERT match those distributions poorly, and that accuracy drops close to chance once annotators split. This report tests two later public NLI checkpoints, DeBERTa-v3 and ModernBERT, with the same metrics. We find that the two models do not move in the same direction. DeBERTa-v3 reaches the highest majority-label accuracy on ChaosNLI-SNLI (79.3%, versus 78.7% for RoBERTa-large) but has the worst Jensen–Shannon distance (JSD) and Kullback–Leibler (KL) divergence of any model we compare. ModernBERT has the lowest JSD and KL divergence, including on MNLI, while remaining below RoBERTa on majority accuracy. Both models stay overconfident on high-disagreement items, and they often pick the same label there. Softening the softmax with temperature scaling cuts JSD without changing any predicted label, so a large part of the distribution error is sharpness rather than a wrong ranking. Majority accuracy and distribution matching remain different skills. Extra NLI training data does not, on its own, improve the second.

1 Introduction

Natural language inference (NLI) asks whether a hypothesis is entailed by a premise, contradicts it, or is neither. SNLI [2] and MNLI [14] treat each pair as having one correct label, usually the majority of a handful of annotators. That setup hides a lot of real disagreement. ChaosNLI [8] re-annotated 1,514 SNLI and 1,599 MNLI-matched development examples with 100 labels each. A fair number of original majority labels do not match the 100-annotator majority, and many items have high label entropy.

Nie et al. [8] compared BERT, RoBERTa, XLNet, BART, and ALBERT against those 100-way distributions. Two results from that paper still matter here. Models that did well on ordinary accuracy were not the ones that matched the human distribution, measured by Jensen–Shannon distance (JSD) and Kullback–Leibler (KL) divergence. Accuracy was high when annotators already agreed and close to chance when they did not. Most remaining errors sat in the low-agreement slice. If later NLI models keep improving majority accuracy, it is not obvious that they also get better at representing disagreement.

We evaluate two encoder checkpoints released after the ChaosNLI study by Nie et al., DeBERTa-v3 and ModernBERT, and compare them with the encoder baselines reported in that work. Both checkpoints were fine-tuned on multiple NLI datasets. This makes them useful for examining whether newer NLI checkpoints differ in how well they match human label distributions. However, neither model was trained directly to predict the 100-annotator distributions used by ChaosNLI. Their different architectures and training setup may lead to different distributional behavior. In particular, the models may differ in how they represent uncertainty and disagreement, even when their majority-label accuracy is similar. We therefore examine whether these differences are also reflected in their ability to match human disagreement, rather than assuming that higher majority-label accuracy necessarily indicates better distributional matching.

We keep the original ChaosNLI metrics: JSD, KL divergence, old accuracy (original dataset label), and new accuracy (100-annotator majority). These metrics allow us to compare the models with the results reported in the original ChaosNLI study. We also split examples by annotation entropy to examine how model performance changes between cases with high and low human disagreement. We compare DeBERTa-v3 and ModernBERT with each other to see whether they make similar predictions on examples with different levels of human disagreement. We then apply temperature scaling when model confidence and human uncertainty come apart, testing whether calibration improves distributional matching without changing the predicted labels.

The rest of the report is organized around three questions:

- How well do these two checkpoints match ChaosNLI human label distributions, relative to the 2020 encoder baselines?
- Does higher majority-label accuracy also mean a better match to human disagreement?
- Does model confidence track human uncertainty, and can a simple post-hoc calibration step improve the distribution scores without changing predicted labels?

2 Related Work

Research on disagreement in NLI has moved from questioning the use of a single gold label toward evaluating and modeling distributions of human judgments. The following sections review this development, from human disagreement and ChaosNLI to approaches for modeling and calibrating human opinion distributions, and finally to recent NLI models and the remaining gap. Together, these studies provide the context for our evaluation.

2.1 Human Disagreement in NLI and ChaosNLI

Human disagreement has been studied in NLI because of the difficulty of assigning a single label to some sentence pairs. Pavlick and Kwiatkowski [9] showed that disagreements between annotators often persist even when more judgments are collected and argued that they cannot simply be treated as annotation noise. They also found that the uncertainty expressed by NLI models does not necessarily correspond to human uncertainty. Their work highlights the need to consider more than a single aggregate label when evaluating NLI models.

ChaosNLI [8] builds on this idea by collecting 100 annotations for selected examples from SNLI, MNLI-matched, and αNLI. The resulting label distributions make it possible to assess how closely model predictions reflect the range of human judgments. The ChaosNLI majority label differs from the original dataset label for 24.97% of SNLI examples and 31.77% of MNLI-matched examples. The level of human disagreement also varies considerably across examples.

Nie et al. [8] compared several NLI models, including BERT, RoBERTa, XLNet, BART, ALBERT, and DistilBERT, using both accuracy and distributional measures such as JSD and KL divergence. DistilBERT achieved comparatively strong KL scores and weak accuracy. The authors interpreted this as evidence that majority classification and distribution matching are not the same skill. They also found that model accuracy decreased as human disagreement increased. These results show why both distributional matching and majority-label accuracy are relevant when evaluating models under human disagreement.

2.2 Modeling and Calibrating Human Disagreement

Later work explored ways to make NLI models better represent human disagreement. Zhou et al. [15] introduced Distributed NLI, where models predict the distribution of human judgments instead of a single label. They showed that methods such as recalibration and distribution estimation can improve the match between model predictions and human opinion distributions. However, the best results remained below the estimated human upper bound.

Meissner et al. [7] changed the training target from a single gold label to a human label distribution. Their results indicated that training on these distributions can reduce JSD on ChaosNLI. Wang et al. [12] took a different approach and focused on calibration. They found that label smoothing and temperature scaling can help model probabilities capture the distribution of human judgments.

Baan et al. [1] studied calibration when human annotators disagree. They argued that ordinary calibration metrics based on a single majority label are misleading when the ground truth itself is a distribution. They proposed measures that take the full human judgment distribution into account.

Lee et al. [5] evaluated instruction-tuned language models on ChaosNLI and found that these models struggled to match human disagreement distributions. For example, GPT-3 did not beat RoBERTa-large on JSD. Their study focuses on generative models and does not evaluate the two encoder checkpoints we use.

Overall, these studies show that majority-label prediction and human-distribution matching are different goals. They also suggest that calibrating model probabilities can improve distributional matching.

2.3 Recent NLI Models and the Remaining Gap

Since the original ChaosNLI study, newer NLI checkpoints have been developed and some have also been evaluated under human disagreement. Tao et al. [11], for example, evaluated DeBERTa-v3-base and RoBERTa-Large on ChaosNLI and compared several post-hoc calibration methods. Their study provides a recent reference for evaluating model confidence when human annotators disagree. However, the DeBERTa-v3-base checkpoint differs from the large multi-dataset DeBERTa-v3 checkpoint used in our experiments, and their study does not include ModernBERT.

The two newer NLI checkpoints evaluated in our study are:

- **DeBERTa-v3** [3, 4]. Fine-tuned on MNLI, FEVER-NLI, ANLI, LingNLI, and WANLI. SNLI was left out of training because of known label issues. Checkpoint: `MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli`.
- **ModernBERT** [10, 13]. Multi-task fine-tuned on a large set of NLI-format datasets, including MNLI, ANLI, WANLI, LingNLI, SICK, and several logic and document NLI tasks. Checkpoint: `tasksource/ModernBERT-large-nli`.

The two checkpoints differ in both architecture and the NLI datasets used for fine-tuning. Both include WANLI (Worker-AI Collaboration for NLI)., which was created through human-AI collaboration to construct challenging NLI examples [6]. Importantly, neither checkpoint was trained to predict the

100-annotator label distributions used by ChaosNLI.

To our knowledge, neither of these checkpoints has been evaluated on ChaosNLI using the original distributional metrics. We also found no ChaosNLI results for either checkpoint in their Hugging Face model cards. Our study extends this line of work by evaluating DeBERTa-v3 and ModernBERT against the full human label distributions in ChaosNLI.

3 Dataset

We use two subsets of ChaosNLI [8]: ChaosNLI-SNLI and ChaosNLI-MNLI-matched. ChaosNLI-SNLI contains 1,514 examples and ChaosNLI-MNLI-matched contains 1,599. Each example has 100 crowd labels, a majority label, a normalized label distribution over entailment, neutral, and contradiction, and the original SNLI or MNLI label. We do not use ChaosNLI-αNLI. The selected checkpoints are 3-way NLI classifiers, not abductive two-choice models.

Human entropy is computed from the 100-label distribution, in bits. We split each subset into tertiles of that entropy and call them low, medium, and high disagreement. This allows us to compare model behavior across different levels of human disagreement.

The 2020 baseline predictions come from the official ChaosNLI release. We recompute JSD, KL, and both accuracies from those files so that our numbers sit on the same protocol as the published scoreboard.

4 Methods

We evaluate DeBERTa-v3 and ModernBERT on ChaosNLI using classification, distributional measures, and qualitative analysis. Their predictions are compared with the human label distributions and the original ChaosNLI baselines. We also analyze model confidence and agreement between the two checkpoints. Finally, temperature scaling is used to test whether adjusting model confidence improves distributional matching without changing the predicted labels.

4.1 Model Inference

We load both models with Hugging Face Transformers and run them in evaluation mode. No weights are updated on ChaosNLI. Premises and hypotheses are tokenized as pairs, with padding and truncation. We do not set a custom maximum length, so each model uses its own tokenizer default (512 for DeBERTa-v3, 8192 for ModernBERT).

For each example, we obtain a probability distribution over the three NLI classes using softmax. The probabilities are aligned to the order (entailment, neutral, contradiction). The predicted label is the class with the highest probability:

$$\hat{y} = \arg\max_i p_i.$$

Confidence is defined as the maximum class probability:

$$c = \max_i p_i.$$

We keep the full probability distribution because it is required for the distributional metrics and uncertainty analysis. For each example, we store the predicted label, probability distribution, and confidence.

4.2 Evaluation Metrics

We first evaluate whether the model prediction matches the reference label using two accuracy measures. Old accuracy compares the model to the original dataset label. New accuracy compares it to the ChaosNLI majority.

Accuracy only considers the predicted label and does not capture the full distribution of human judgments. We therefore use KL divergence and JSD to compare the model probability distribution with the human label distribution.

KL divergence measures the difference between two probability distributions. In our evaluation, it is computed as

$$\mathrm{KL}(p_{\mathrm{human}} \| p_{\mathrm{model}}).$$

Here, the human distribution is treated as the reference and the model distribution is evaluated against it. We report KL in nats. Smaller KL values indicate greater similarity between the two distributions.

JSD provides a symmetric comparison of the human and model distributions. We first compute their average distribution, followed by the Jensen–Shannon divergence (JSDiv). We use then SciPy's JSD, which is obtained as the square root of JSDiv and uses natural log. This matches the ChaosNLI evaluation code:

$$m = \frac{1}{2}\left(p_{\mathrm{human}} + p_{\mathrm{model}}\right).$$

$$\mathrm{JSDiv} = \frac{1}{2}\mathrm{KL}(p_{\mathrm{human}} \| m) + \frac{1}{2}\mathrm{KL}(p_{\mathrm{model}} \| m).$$

$$\mathrm{JSD}(p, q) = \sqrt{\mathrm{JSDiv}(p_{\mathrm{human}}, p_{\mathrm{model}})}.$$

Lower JSD values mean that the model is closer to the human distribution.

We further analyze model uncertainty using confidence and entropy. As described in Section 4.1, model confidence is the maximum predicted class probability and captures how strongly the model favors its predicted label. Model entropy measures how the probability is distributed across the three NLI classes. It is computed in bits as

$$H(p) = -\sum_i p_i \log_2 p_i.$$

Higher entropy represents a more evenly spread prediction and greater model uncertainty. We calculate the mean confidence and mean model entropy across the examples.

In addition, we compute the Spearman correlation between model confidence and human entropy, and between model entropy and human entropy. These correlations examine whether model confidence or uncertainty changes in the same direction as human disagreement.

4.3 Baselines and Model–Model Comparison

To compare our results with the original ChaosNLI evaluation, we use the prediction files provided for BERT-large, RoBERTa-large, XLNet-large, BART-large, and ALBERT-xxlarge. Their logits are converted to probability distributions with softmax. We recompute the JSD, KL divergence, old accuracy, and new accuracy from these predictions. Recomputed SNLI and MNLI scores match the published table to four decimals.

For each example, we record whether DeBERTa-v3 and ModernBERT predict the same label, and whether that shared label matches the human majority. We perform this comparison separately for the low-, medium-, and high-disagreement groups defined by human entropy. We then compare model consistency across these groups and check whether they make the same errors on ambiguous examples.

This comparison is not in the original ChaosNLI paper. The point is to see whether two independently trained modern NLI models fall back on the same convention when humans do not agree.

4.4 Temperature scaling

The saved prediction files store softmax probabilities, not raw logits, so we recover logits as $\log p$ before searching for a temperature T. We draw a random 50/50 split (seed 0), search $T \in [0.5, 5.0]$ in 19 equal steps to minimize JSD on one half, and score the other half. Argmax is invariant to $T > 0$, so accuracy cannot change. We also search for a T that matches mean model entropy to mean human entropy, without using labels.

4.5 Qualitative Analysis

The selection of examples for the qualitative analysis is not random. We use targeted sampling based on specific properties of the model predictions and human disagreement. We compare the predictions of DeBERTa-v3 and ModernBERT and select examples that meet particular criteria.

Before describing these criteria, we clarify two terms. The SNLI and MNLI-matched datasets provide one label for each example. We refer to this as the *original dataset label*. ChaosNLI provides a label distribution based on 100 annotations collected from crowd workers. The label with the highest proportion in this distribution is referred to as the *human majority label*. The *original dataset label* and the *human majority label* do not always agree. This disagreement is used as a selection criterion in two cases described below.

High human entropy identifies examples with substantial annotator disagreement. These examples are useful for qualitative analysis, as they capture cases where human judgments diverge significantly. High model confidence highlights cases where a model remains highly certain despite this disagreement. This allows us to inspect examples where the model distribution is much sharper than the human distribution.

Specifically, we use four selection criteria:

- high-entropy examples where DeBERTa-v3 and ModernBERT predict the same label with high confidence, but this label differs from the *human majority label*;
- high-entropy examples where DeBERTa-v3 and ModernBERT predict different labels;
- examples where the *original dataset label* differs from the *human majority label*, but both models

predict the *human majority label*;

- examples where the *original dataset label* differs from the *human majority label*, and both models predict the *original dataset label*.

For the first two criteria, we use a human entropy threshold of 1.0. For the last two criteria, we use a threshold of 0.8. The selected examples are manually inspected. Representative cases are then chosen to identify recurring patterns of model–human disagreement.

5 Results

We begin by comparing the two models with the original ChaosNLI baselines. We then look more closely at their behavior under human disagreement, including their confidence, agreement, and distributional predictions. The final analyses consider temperature scaling and qualitative examples.

5.1 Overall comparison

Table 1 and Table 2 report official ChaosNLI metrics. Lower JSD and KL are better. Higher accuracy is better.

Table 1: ChaosNLI-SNLI ($n = 1514$). 2020 rows are recomputed from the official prediction files.

Model	JSD ↓	KL ↓	Old acc. ↑	New acc. ↑
BERT-large (2020)	0.230	0.502	0.727	0.738
RoBERTa-large (2020)	0.221	0.494	0.749	0.787
XLNet-large (2020)	0.226	0.505	0.743	0.781
BART-large (2020)	0.220	0.471	0.742	0.783
ALBERT-xxlarge (2020)	0.235	0.534	0.715	0.781
DeBERTa-v3 (ours)	0.278	0.858	0.709	0.793
ModernBERT (ours)	0.214	0.378	0.708	0.768

Table 2: ChaosNLI-MNLI-matched ($n = 1599$).

Model	JSD ↓	KL ↓	Old acc. ↑	New acc. ↑
BERT-large (2020)	0.315	0.845	0.612	0.569
RoBERTa-large (2020)	0.311	0.870	0.674	0.635
XLNet-large (2020)	0.312	0.882	0.674	0.619
BART-large (2020)	0.317	0.885	0.664	0.592
ALBERT-xxlarge (2020)	0.316	0.862	0.649	0.590
DeBERTa-v3 (ours)	0.364	1.465	0.695	0.634
ModernBERT (ours)	0.285	0.638	0.671	0.624

On SNLI, DeBERTa-v3 has the highest new accuracy (0.793) and the worst JSD and KL. Its old SNLI accuracy (0.709) is below RoBERTa-large (0.749). That split fits the model card: SNLI was not used in training. The checkpoint lines up better with the 100-annotator majority than with the original five-annotator labels, and still produces very peaked distributions (mean confidence 0.947, mean model entropy 0.21 bits, against human entropy 0.80).

ModernBERT has the best SNLI JSD (0.214) and KL (0.378), better than BART-large, which was the previous JSD leader in this set. Its new accuracy (0.768) is below RoBERTa-large.

On MNLI the 2020 models already sat near the chance JSD band (about 0.31). ModernBERT is the first model in Table 2 to move clearly below that band (JSD 0.285, KL 0.638). DeBERTa-v3 improves old MNLI accuracy (0.695 vs. 0.674 for RoBERTa) and makes KL almost twice as large (1.465 vs. 0.870). New accuracy is essentially tied with RoBERTa (0.634 vs. 0.635).

Higher majority accuracy and better distribution match do not arrive together on these two checkpoints.

5.2 Entropy bins

Table 3 shows new accuracy by human-entropy tertile. Figure 1 and Figure 2 plot the SNLI split. Figure 4 and Figure 5 show the same split on MNLI.

Table 3: New accuracy by human-disagreement tertile.

Split	Low	Medium	High
SNLI DeBERTa-v3	0.948	0.820	0.612
SNLI ModernBERT	0.926	0.781	0.598
SNLI RoBERTa-large	0.952	0.806	0.602
MNLI DeBERTa-v3	0.705	0.619	0.576
MNLI ModernBERT	0.711	0.584	0.576
MNLI RoBERTa-large	0.704	0.636	0.567

On high-disagreement items, all three models are around 60% on SNLI and 57% on MNLI. The low-entropy SNLI bin is nearly solved (about 95%). That is the same shape Nie et al. reported. The newer encoders do not move the high-entropy slice.

Mean confidence on SNLI stays high for DeBERTa-v3 even when humans disagree (0.978 / 0.946 / 0.918 across the three bins). ModernBERT drops more (0.924 / 0.850 / 0.795), which is one reason its JSD is lower. On MNLI, Spearman ρ between DeBERTa confidence and human entropy is -0.05. For ModernBERT it is -0.17. Neither softmax is a reliable signal of MNLI ambiguity.

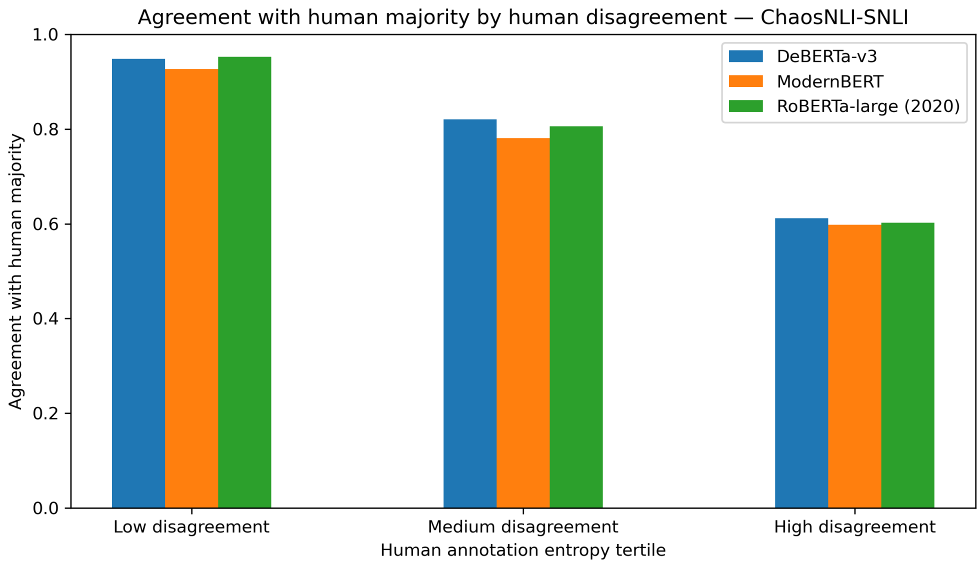

Figure 1: Agreement with the 100-annotator majority by human-disagreement tertile on ChaosNLI-SNLI.

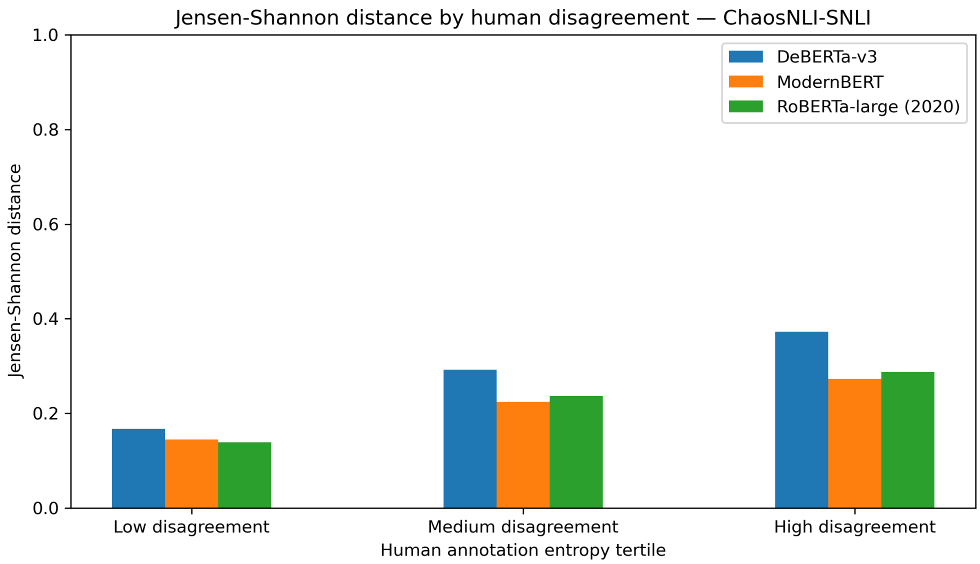

Figure 2: Jensen–Shannon distance by human-disagreement tertile on ChaosNLI-SNLI.

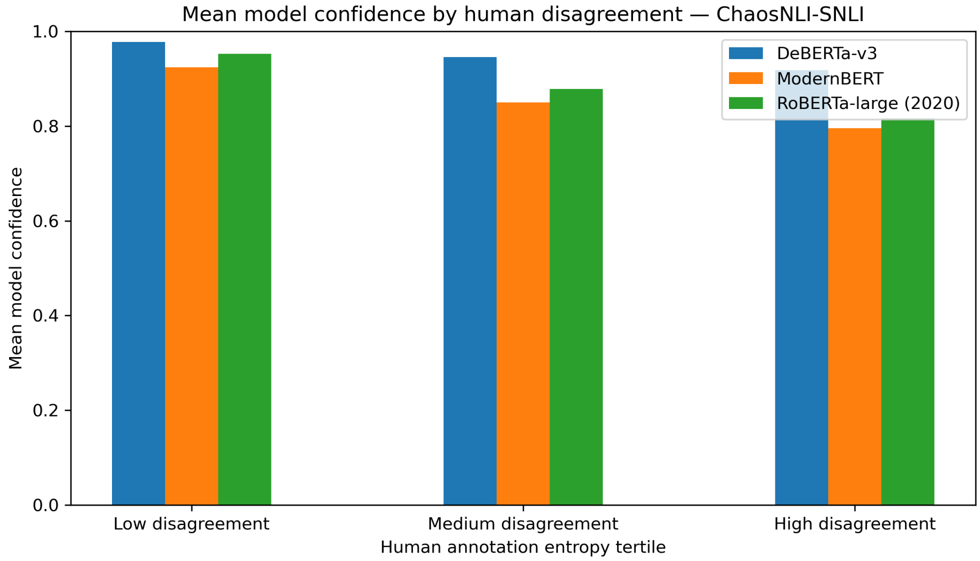

Figure 3: Mean model confidence by human-disagreement tertile on ChaosNLI-SNLI.

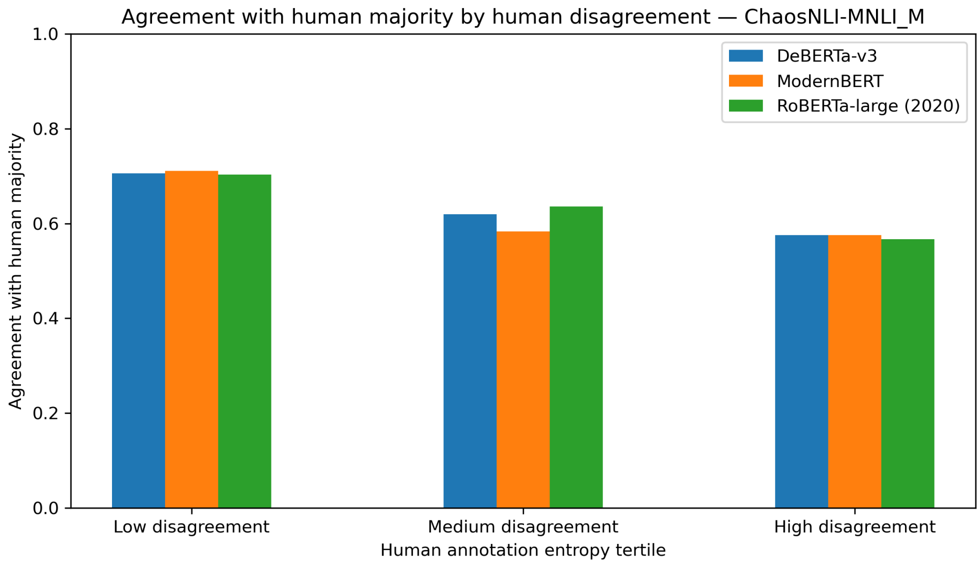

Figure 4: Agreement with the 100-annotator majority by human-disagreement tertile on ChaosNLI-MNLI-matched.

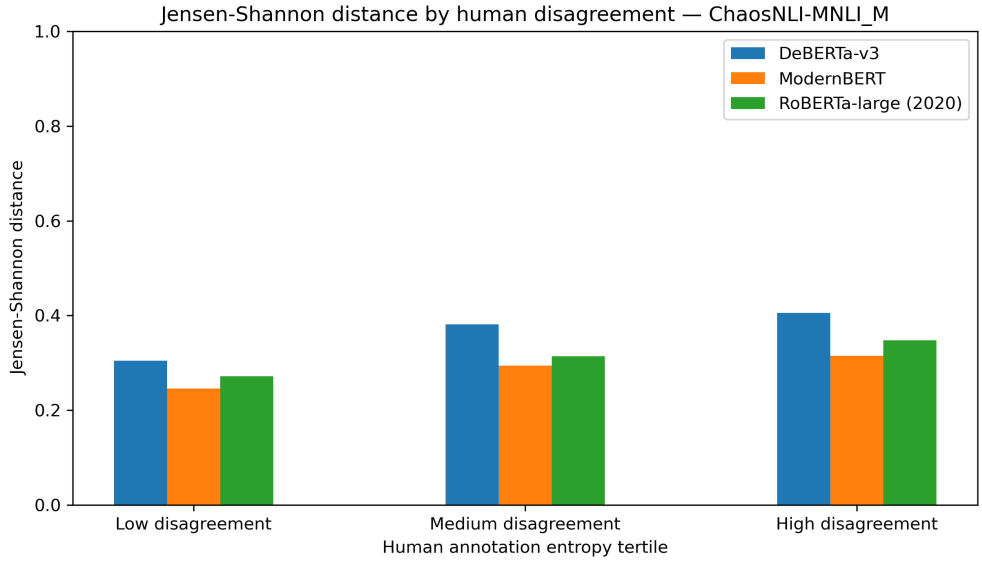

Figure 5: Jensen–Shannon distance by human-disagreement tertile on ChaosNLI-MNLI-matched.

5.3 Agreement between the two modern models

Table 4 and Table 5 compare DeBERTa-v3 with ModernBERT. Figure 6 shows the SNLI curves.

Table 4: DeBERTa-v3 and ModernBERT on ChaosNLI-SNLI, by entropy tertile.

Tertile	Models agree	Both agree, both miss majority	Mean DeBERTa conf.
Low	0.913	0.018	0.978
Medium	0.794	0.095	0.946
High	0.679	0.224	0.918

Table 5: Same comparison on ChaosNLI-MNLI.

Tertile	Models agree	Both agree, both miss majority	Mean DeBERTa conf.
Low	0.803	0.191	0.930
Medium	0.788	0.291	0.941
High	0.705	0.263	0.915

On high-entropy SNLI the two models still pick the same label 68% of the time. In 22% of that tertile they agree with each other and not with the human majority. On MNLI they jointly miss the majority even in the low-entropy bin (19%). They share MNLI, ANLI, WANLI, and LingNLI as training sources, and they share the same three-way label set. The shared errors look more like a common NLI convention than like two independent readings of the same ambiguity.

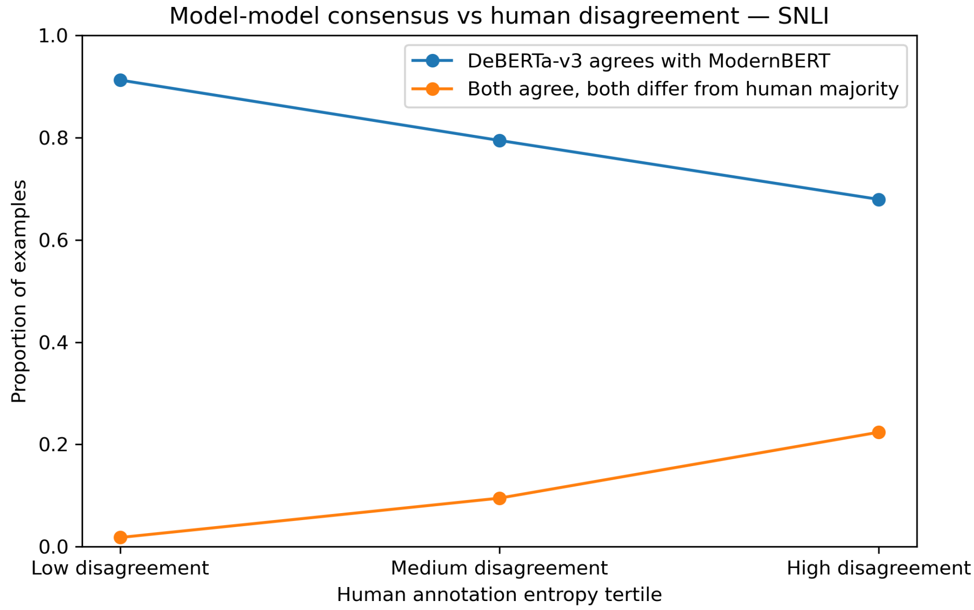

Figure 6: Model–model consensus versus human disagreement on ChaosNLI-SNLI.

5.4 Temperature scaling

Table 6 reports held-out temperature scaling.

Table 6: Temperature T fit on a random half of each split, scored on the other half. Accuracy does not change.

Model / split	T	Held-out JSD before / after	Held-out KL before / after
DeBERTa SNLI	2.75	0.276 / 0.187	0.848 / 0.195
ModernBERT SNLI	1.75	0.217 / 0.188	0.389 / 0.206
DeBERTa MNLI	4.50	0.358 / 0.206	1.430 / 0.211
ModernBERT MNLI	2.75	0.278 / 0.192	0.603 / 0.183

Temperature scaling reduces both JSD and KL divergence for all four model–dataset combinations. After softening, both models sit near JSD 0.19. The largest change occurs for DeBERTa-v3 on MNLI, where JSD decreases from 0.358 to 0.206 and KL from 1.430 to 0.211. This setting also requires the highest temperature ($T = 4.5$), indicating that the original probability distribution is particularly sharp. DeBERTa-v3 on SNLI requires a smaller but still substantial temperature of $T = 2.75$. ModernBERT needs less softening, with $T = 1.75$ on SNLI and $T = 2.75$ on MNLI.

The entropy-matching experiment gives a similar result for DeBERTa-v3 on SNLI. Without using the labels, matching the mean model entropy to the mean human entropy gives $T = 2.6$ and reduces JSD from 0.278 to 0.187. This is close to the JSD obtained with the temperature selected directly on the held-out JSD. The result suggests that the main difference between the original model distribution and the human distribution is partly related to prediction sharpness.

However, temperature scaling cannot reproduce the full structure of human disagreement. Because a single temperature is applied to every example, it only modulates overall prediction sharpness. Human disagreement varies from example to example. One item may have a distribution close to 50/50 between entailment and contradiction, while another may have a clear 95% majority for entailment. Both examples are softened by the same temperature, even though their underlying uncertainty is different.

This limitation is important when interpreting the improvement in JSD and KL divergence. Temperature scaling can reduce excessive confidence and bring the model distributions closer to the human distributions, but it does not model item-specific disagreement. The experiment consequently suggests that part of the distribution mismatch comes from overly sharp predictions.

5.5 Qualitative examples

The selected examples reveal several recurring patterns in model–human disagreement. Four patterns are discussed below.

- **Lexical mismatch that humans treat as scene description**

  One recurring pattern arises when the premise and hypothesis describe different actions or details of the same scene. The models can interpret these differences more strictly, while human annotators may treat them as compatible scene details.

  Premise: "A man in a dark suit talks with a middle-aged man and woman on a stage in front of a large logo of the letter 'D' with the numeral five within it."

  Hypothesis: "A couple of people are sitting in a room."

  Human Distribution: entailment 18%, neutral 58%, contradiction 24%.

  DeBERTa-v3 and ModernBERT: contradiction at $\geq 98\%$ confidence.

  Both models treat the difference between "talks" and "sitting" as a contradiction and assign almost all probability to contradiction. In contrast, the human distribution is dominated by neutral, suggesting that many annotators do not consider the two actions incompatible. A person can talk with others while sitting, so the hypothesis can be interpreted as a compatible description of the scene.

  This example illustrates how a lexical mismatch between the described actions can lead to a much sharper model prediction than the human distribution.

- **Pragmatic inference versus strict entailment**

  A second pattern concerns cases where human annotators accept a plausible real-world inference as entailment, even though the hypothesis is not directly supported by the premise.

  Premise: "A boy in jeans, black t-shirt, and blue cap on a bike, leaps over steps."

  Hypothesis: "A boy is practicing bike tricks."

  Human Distribution: entailment 45%, neutral 45%, contradiction 10%.

  DeBERTa-v3 and ModernBERT: neutral at $\geq 95\%$ confidence.

  The human distribution is almost evenly split between entailment and neutral, with a slight majority for entailment. Many annotators infer that the boy is practicing bike tricks, even though this is not explicitly stated in the premise. Both DeBERTa-v3 and ModernBERT instead predict neutral, matching the original SNLI dataset label. They favor a stricter interpretation and require more direct evidence for entailment.

  This example suggests that human annotators may accept a pragmatic inference that our models do not treat as sufficient for entailment.

- **Different judgments for similar statements** Human disagreement can also occur when the premise and hypothesis are very similar.

Premise: “Closed on the Sabbath.”

Hypothesis: “Sabbath is closed.”

Human Distribution: entailment 37%, neutral 24%, contradiction 39%.

DeBERTa-v3 and ModernBERT: entailment at approximately 98% probability.

Both models produce highly concentrated distributions, with approximately 98% of the probability on entailment. The premise and hypothesis are closely related in wording, which may explain why the two models arrive at the same prediction. At the same time, the human annotations are divided between entailment and contradiction.

This case demonstrates that human judgments can differ markedly even when the models make the same highly confident prediction.

- **Different representations of uncertainty** When the models disagree, ModernBERT is usually less peaked than DeBERTa-v3. Its probability is distributed more evenly across alternative labels, whereas DeBERTa-v3 often concentrates most of its probability on a single class.

  Premise: “A skateboard performs a trick oof a quarter pipe.”

  Hypothesis: “A skateboard does a trick without a skateboarder.”

  Human Distribution: entailment 14%, neutral 49%, contradiction 37%.

  DeBERTa-v3: neutral at 99% confidence, JSD $= 0.45$.

  ModernBERT: entailment 11%, neutral 41%, contradiction 48%, JSD $= 0.08$.

  The premise describes a skateboard performing a trick. The hypothesis adds the detail that no skateboarder is present. The human annotators are largely split between neutral and contradiction. DeBERTa-v3 places almost all probability on neutral, while ModernBERT distributes its probability mainly between neutral and contradiction with the highest probability on contradiction. ModernBERT has a much lower JSD and is closer to the human distribution, although contradiction is not the human majority label.

  This case shows that a less concentrated probability distribution can better represent human disagreement even when the predicted label does not match the human majority label.

The qualitative examples cover several forms of model–human disagreement. DeBERTa-v3 and ModernBERT can remain highly confident even with variation in the human annotations. They can also apply stricter interpretations than some annotators. In addition, the two models differ in how they represent alternative labels. These observations support the use of distributional measures alongside majority-label accuracy when evaluating model alignment with human judgments.

6 Discussion

Both checkpoints are 3-way classifiers trained with cross-entropy on one gold label. That loss is minimized by putting almost all probability on one class. ChaosNLI scores the full softmax against a 100-person distribution that is often spread across two labels. If humans put 20% on contradiction and the model puts 0.1%, KL becomes large even when the model’s top label is the majority. DeBERTa-v3 can therefore be the strongest majority classifier in the SNLI table and the weakest distribution model at the same time. Nie et al. already saw a version of this with DistilBERT. The same split is more extreme on the Laurer checkpoint.

DeBERTa-v3 was trained on a large hard-label mix (about 885k pairs). Extra one-hot NLI data helps the ranking of the majority class and also trains very certain logits. Mean SNLI confidence is 0.95. On MNLI, confidence barely moves with human entropy. Temperature scaling is a direct test of that account: $T = 2.75$ on SNLI and $T = 4.5$ on MNLI cut JSD roughly in half without changing any predicted label. The ranking was often already right. The distribution was too sharp.

ModernBERT's mix is broader, with many NLI-format tasks under multi-task fine-tuning. It never becomes as peaked: mean SNLI confidence 0.86, mean model entropy 0.53 bits rather than 0.21. A softer distribution is closer to a spread-out human distribution, which improves JSD and KL even when the argmax is not better. WANLI and related sets are built from worker disagreement, so the model sees more label variation than a classic MNLI-only encoder, still under hard labels. That is not the same as learning the ChaosNLI 100-way target, but it can limit collapse onto a single convention. On the skateboard example, ModernBERT does not resolve the ambiguity better in any deep sense. It is less willing to zero out the other plausible class.

High-entropy items look underspecified rather than merely difficult: verb mismatch, extra inferred detail, near-paraphrase. A stronger encoder can learn the majority convention on low-entropy items (DeBERTa reaches 94.8% there on SNLI). Hard-label training has no target of the form "45% of people will call this entailment," so the high-disagreement slice stays near the 2020 numbers.

The two modern models agree with each other more than they agree with divided humans because they share a label space and overlapping NLI sources. They learn similar heuristics: lexical overlap, conservative neutrality on extra details, sharp contradiction on mismatched verbs. Humans split for other reasons. That looks like a shared training convention, not two systems recovering the same human uncertainty.

We did not retrain DeBERTa on ModernBERT's mix, or the reverse. The account above is based on the metrics, the model cards, and the temperature results, not on a causal ablation. It is consistent with [8, 7, 1].

7 Limitations

We evaluate frozen checkpoints. Soft-label training on ChaosNLI is already known to help [15] and is a different experiment. We omit αNLI. Temperature scaling uses a random split of ChaosNLI itself, so it is a diagnostic rather than a calibrator one would ship. A single global temperature only adjusts prediction sharpness and cannot capture item-specific human disagreement. Because we stored softmax probabilities rather than logits, the temperature search is run on $\log p$, which is an approximation to the original logits.

JSD and KL follow the official ChaosNLI implementation so that 2020 rows remain comparable. The discussion of training data is inferred from public model cards, not from a controlled re-train.

The evaluation is limited to two recent NLI checkpoints and two ChaosNLI subsets, so the findings may not generalize to other models or datasets. The qualitative analysis is based on a small number of targeted examples. The provided patterns here are not exhaustive and do not form a systematic taxonomy of model–human disagreement.

8 Conclusion

We scored two later NLI checkpoints on ChaosNLI with the original JSD, KL, old accuracy, and new accuracy protocol. DeBERTa-v3 improves majority-label accuracy on SNLI and old accuracy on MNLI, and it is worse than the 2020 encoders on distribution match. ModernBERT improves JSD and KL, including a clear MNLI JSD gain, without beating RoBERTa-large on majority accuracy. High-disagreement accuracy stays near 60%. The two models often pick the same label when annotators do not. Temperature scaling shows that much of the JSD gap is excess confidence. Encoder NLI progress since 2020 did not, by itself, close the disagreement problem posed by ChaosNLI.

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ChaosNLI data and the 2020 baseline predictions are from Nie, Zhou, and Bansal. The two checkpoints are public Hugging Face models from Laurer et al. and from tasksource. This report was prepared as a seminar project.

A Reproducibility

Predictions, metrics, and plots are under `results/`. From the project root:

```
python src/evaluate_models.py deberta snli
python src/evaluate_models.py deberta mnli_m
python src/evaluate_models.py modernbert snli
python src/evaluate_models.py modernbert mnli_m
python src/analyze_findings.py
```

The four evaluation jobs write JSON prediction files. The analysis script then recomputes the official ChaosNLI metrics against the 2020 baselines, builds the entropy-bin tables and plots, runs temperature scaling, and writes `results/findings_summary.json`. The report source is `report/nli_disagreement.tex`. Compile with `pdflatex` from the `report/` directory.

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